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Former Glove

AI Training Gloves

Diplomarbeit

Schulautonomer Schwerpunkt

Coding & AI

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Abstract (Kevin)

Effective strength training relies heavily on correct technique, yet many athletes perform exercises with poor form, increasing the risk of injury and reducing training efficiency. While professional coaching provides a solution, it is often cost-prohibitive or inaccessible for the average gym-goer.

To address this problem, we developed "Former Glove", a smart wearable system that acts as a digital coach. The project consists of sensor-equipped gloves that track hand movements during weightlifting. Using intelligent software, the system analyzes the exercise in real-time to detect mistakes and count repetitions automatically. This feedback is wirelessly transmitted to a mobile app, allowing the user to see exactly how well they performed each set.

By offering immediate, automated guidance, "Former Glove" provides the benefits of professional coaching without the high cost, helping users to train more safely and effectively.

Kurzfassung (Kevin)

Effektives Krafttraining basiert maßgeblich auf korrekter Technik. Viele Athleten führen Übungen jedoch mit fehlerhafter Form aus, was das Verletzungsrisiko erhöht und die Trainingseffizienz mindert. Professionelles Coaching bietet zwar eine Lösung, ist aber für den durchschnittlichen Fitnessstudiobesucher oft zu teuer oder nicht zugänglich.

Um dieses Problem zu lösen, haben wir „Former Glove“ entwickelt, ein intelligentes tragbares System, das als digitaler Trainer fungiert. Das Projekt besteht aus mit Sensoren ausgestatteten Handschuhen, die die Handbewegungen beim Gewichtheben erfassen. Mithilfe intelligenter Software analysiert das System die Übung in Echtzeit, erkennt Fehler und zählt die Wiederholungen automatisch. Dieses Feedback wird drahtlos an eine mobile App übertragen, sodass der Nutzer genau sehen kann, wie gut er jeden Satz ausgeführt hat.

Durch die sofortige, automatisierte Anleitung bietet „Former Glove“ die Vorteile professionellen Coachings ohne die hohen Kosten und hilft Nutzern, sicherer und effektiver zu trainieren.

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1 Introduction (Kevin)

Weight training has become one of the most popular forms of exercise worldwide. Whether for aesthetic reasons, rehabilitation, or general health, millions of people lift weights every day. However, simply moving a weight from point A to point B is not enough. The key to success lies in the quality of the movement.

1.1 Problem Setting (Kevin)

Imagine a beginner walking into a gym for the first time and trying to do bicep curls with no prior experience. Instead of using only their arm muscles, they swing their entire upper body to lift the weight. To a trained professional, the mistake is obvious. To a beginner, however, it feels like a success because the weight moved up.

This scenario illustrates a fundamental problem in individual sports: The lack of immediate and objective feedback.

Performing exercises with incorrect form has two major negative consequences. First, it drastically reduces the effectiveness of the training, as the target muscles are not properly engaged. Second, and more importantly, it significantly increases the risk of acute and chronic injuries.

While professional athletes rely on personal trainers to monitor their form, this solution is not scalable for the general public. Personal trainers are expensive, often costing significantly more per hour than the monthly gym membership itself, and they cannot be present for every single workout session a person does. Consequently, a vast number of gym-goers are left to train unassisted, unaware of their mistakes, and prone to injury.

1.2 Problem Statement (Kevin)

The central problem is the absence of an accessible, technological solution that can substitute the help of a professional trainer.

While wearable technology is widely available in the sector of endurance sports (tracking heart rate, steps, or distance), solutions for strength training remain largely underdeveloped. Existing fitness trackers primarily focus on quantitative data (e.g., "How many reps did I do?"), but they fail to capture qualitative data (e.g., "Was the movement performed correctly?").

Furthermore, stationary alternatives such as camera-based systems are often impractical for gym environments due to the lack of mobility and privacy concerns. Consequently, there is currently no mobile, affordable, and privacy-compliant system available that can autonomously analyze biomechanical movement patterns in real-time and provide corrective feedback to the user.

1.3 Solution (Kevin)

To solve this problem, we invented the "Former Glove" system. The core idea is to replace the eyes of a personal trainer with a smart wearable device that is affordable, always available, and objective.

The solution consists of two main components:

1. **The Former Glove:** A specialized fitness glove equipped with sensitive motion sensors. These sensors act as the system's "sense of touch", recording the precise movement patterns of the user’s hands in a three-dimensional space.
2. **The Former App:** A mobile application that acts as the user interface and "digital coach”, using Bluetooth to connect to the gloves.

Unlike simple fitness trackers that only count steps or repetitions, the Former Glove system understands the quality of a movement. Using artificial intelligence, the system analyzes the

sensor data in real-time. It determines whether an exercise (like a bicep curl) was performed cleanly or with momentum.

The contribution of this work is a fully functional prototype that connects the physical and digital worlds. It allows a user to define a workout, perform it wearing the gloves, and receive immediate feedback on their smartphone regarding repetition counts and execution quality. This allows the user to correct their form instantly, ensuring a safer and more effective workout without the need for a human instructor.

1.4 Structure of this Thesis

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2 Technological Foundation (Kevin)

The implementation of a smart tracking system like the Former Glove requires the integration of different technologies: embedded hardware, artificial intelligence, mobile software and wireless communication protocols.

This Chapter provides the knowledge needed to understand the design and implementation strategies described later in this thesis. It introduces the individual hardware components used for motion tracking, the software frameworks used to train the artificial intelligence Model and to build the smartphone app and the properties of the Bluetooth Low Energy protocol which enables the wireless communication of data between devices.

2.1 Arduino Nicla Sense ME (Daniel)

The Nicla Sense ME is a microcontroller by Arduino. Its miniscule size is practical for integrating it into smaller devices without creating bulk. It is equipped with various sensors provided by Bosch, including an accelerometer and a gyroscope, which we will be using to track the movement of our gloves. It is also specifically designed for running complex AI-based code, which makes it a perfect fit for this project. Two of them will are used, one for each glove.

2.2 Edge Impulse (Daniel)

Edge Impulse is a development platform for embedded machine learning. It enables the collection, preprocessing, and labelling of sensor data, as well as the training and optimization of AI models directly for microcontrollers. It will process the movement data collected and help creating a library which can be directly run on the microcontroller.

2.3 Smartphone App (Kevin)

The user interface (UI) and the logic for workout management are hosted on a mobile application. The "Former" app was developed using the Flutter framework in combination with the Dart programming language and the Android Studio integrated development environment (IDE).

2.3.1 Flutter Framework (Kevin)

Flutter is an open-source UI software development kit (SDK) developed by Google. It allows developers to build applications for mobile, web, and desktop from a single codebase. Unlike traditional development (which would require Kotlin for Android and Swift for iOS), Flutter uses its own rendering engine to draw widgets. This makes sure that the user has a consistent visual appearance across different devices. In Flutter, the entire UI is built by nesting widgets (e.g., buttons, text, layout containers) into a tree structure. This approach is essential for the "Former" app to dynamically rebuild widgets when new sensor data arrives via Bluetooth.

2.3.2 The Dart programming language (Kevin)

The Flutter SDK uses the Dart programming language also developed by Google. Dart is optimized for object-oriented programming. Dart is particularly well suited for the development of the app because of two main features:

- Just-in-Time (JIT) Compilation: This enables the "Hot Reload" feature during development, allowing code changes to be reflected instantly in the running app without losing the current state (e.g., a simplified Bluetooth connection).
- Asynchronous Programming: Dart supports asynchronous operations using Future and Stream objects. This is critical for handling Bluetooth Low Energy (BLE) events, where data packets from the glove arrive asynchronously and must be processed without freezing the user interface.

2.3.3 Android Studio (Kevin)

The development environment used is Android Studio, the official IDE for Android development. It provides the necessary tools for code editing and debugging.

Although Android Studio includes an integrated device emulator for virtual testing, this project relies heavily on the Android Debugging Bridge (ADB), which acts as an interface between the development environment and the physical device. Since standard emulators don't support the use of Bluetooth connections, uploading the application to a smartphone was essential for testing the communication with the Former Glove.

2.4 Communication Protocols (Bluetooth Low Energy) (Kevin)

For the wireless data transmission between the wearable hardware and the mobile application, Bluetooth Low Energy (BLE) was selected as the communication standard. BLE is specifically designed for applications requiring low power consumption and sporadic data transfer, making it the industry standard for fitness wearables.

Unlike classic Bluetooth, which is optimized for continuous data streaming (e.g., audio), BLE remains in sleep mode for the majority of the time and only wakes up to transmit short bursts of data. This characteristic is essential for the Former Glove, as it allows the Nicla Sense ME to operate with a minimal battery consumption while maintaining a connection over the duration of a workout session.

2.4.1 Generic Access Profile (GAP)

The Generic Access Profile (GAP) defines how devices discover each other and establish a connection. In this project, the system architecture assigns specific roles to the hardware components:

- **Peripheral (The Glove):** The Nicla Sense ME acts as the Peripheral. It is a low-power device that "advertises" its presence by broadcasting small data packets containing its name ("LeftGlove") and available services. It waits for a connection request.
- **Central (The Smartphone):** The mobile device running the Flutter application acts as the Central. It has higher processing power and battery capacity. Its role is to scan for advertising packets, initiate the connection, and manage the communication timing.

2.4.2 Generic Attribute Profile (GATT)

Once a connection is established via GAP, the **Generic Attribute Profile (GATT)** defines how data is structured and exchanged. GATT organizes data hierarchically into **Services** and **Characteristics**:

1. **Services:** A Service is a logical collection of data and behaviors to accomplish a specific function. For this project, a custom "Training Service" was defined and identified by a unique UUID (Universally Unique Identifier).
2. **Characteristics:** Characteristics are the actual data containers within a Service. They hold values (e.g., the repetition count or the sensor feedback) and properties that define how this value can be accessed (Read, Write, or Notify).
3. **UUIDs:** To ensure that the app communicates with the correct data points, both the Service and its Characteristics are addressed via predefined UUIDs known to both the firmware and the app software.

2.4.3 Data Transmission Mechanism: Notifications

Efficient data transfer is critical for real time feedback. Standard polling (where the app constantly asks "Do you have new data?") is inefficient and increases latency. Instead, this project utilizes the notification mechanism provided by BLE. The smartphone "subscribes" to the specific characteristic of the glove. Whenever the glove's internal logic detects a new event (e.g., a completed repetition), it actively pushes the update to the smartphone. This event-driven architecture ensures that the application interface updates instantly while minimizing the Bluetooth usage of the microcontroller.

3 System Design

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3.1 Data Flow Diagram (Daniel)

The final project includes three components, all which have to communicate with each other periodically. Firstly we have the right glove, which only has to send its sensor values to the left glove so they can be evaluated together. The right glove is a server and the left glove connects to it as a client via Bluetooth Low Energy (BLE). The left glove is responsible for evaluating the combined sensor time series data using a different AI model for form and repetition detection depending on the selected exercise. It serves both as a BLE client and server, as the mobile app is connected to it over BLE as a client. While the right glove sends both signals for the end of a rep and the evaluated label, the app only has to communicate whenever the user starts a new set of exercises, including the selected exercise, weight and repetitions.

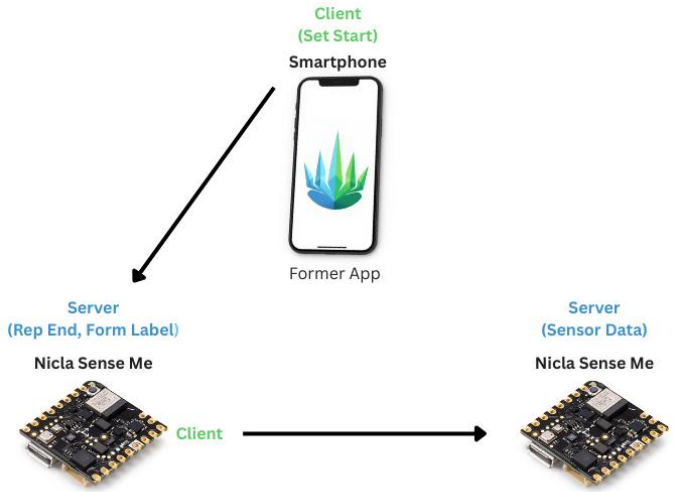


Figure 1: Data Flow Diagram

3.2 Hardware Design (Daniel)

The hardware design is a challenging problem, as training gloves require free movement and too much clutter would hinder the user in exercising. Thus, the best solution that minimizes the size of both the microcontroller's and battery's housing is modelling a mounting rail. The housings are connected to the counterpart of the rail's shape, which makes it possible to slide them together without the need for any screws or clips. The mounting rail has small holes so it can be sewn to the glove.

For the battery, a smaller model with 250mAh is used on both gloves. To charge the microcontrollers, it can easily be connected to the battery pins of the Nicla Sense ME by connecting a plug directly leading to the battery and a socket on the microcontroller's side. The battery will automatically be charged when the Nicla Sense ME is connected via micro-USB cable.

3.3 Smartphone App

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4 Implementation

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4.1 Glove Firmware Logic (Daniel)

The code for the microcontrollers is managed in PlatformIO using normal Arduino code. As multiple main files will be used depending on the current stage in development, every main has its own framework in the initialization file, which all have a similar structure that filters them by name:

```
[env:leftGlove]
platform = nordicnrf52
board = nicla_sense_me
framework = arduino
build_src_filter = +<*.h> +<main-${PIOENV}.cpp>
lib_ldf_mode = deep+
lib_deps =
    arduino-libraries/ArduinoBLE@^1.4.1
    arduino-libraries/Arduino_BHY2@^1.0.8
```

4.1.1 Fetch Data

Because repetitions can’t be detected until a full dataset is created, the original data has to be collected manually. The right glove has to advertise itself:

```
BLE.begin();
  BLE.setLocalName("RightGlove");
```

```
  BLE.setAdvertisedService(sensorValues);
  sensorValues.addCharacteristic(movementChar);
  BLE.addService(sensorValues);
  BLE.advertise();
```

The sensor data also has to be collected periodically and written on the characteristic so the left Glove can access it. The easiest way to do it is to copy the values into the buffer memory:

```
void sendSensorData(){
  float ax = accel.x(), ay = accel.y(), az = accel.z();
  float gx = gyro.x(), gy = gyro.y(), gz = gyro.z();

  byte buffer[24];
  memcpy(&buffer[0], &ax, 4);
  memcpy(&buffer[4], &ay, 4);
  memcpy(&buffer[8], &az, 4);
  memcpy(&buffer[12], &gx, 4);
  memcpy(&buffer[16], &gy, 4);
  memcpy(&buffer[20], &gz, 4);

  movementChar.writeValue(buffer, 24, true);
}
```

In the code of the left glove, the connection to the right glove is handled by searching for the device by name:

```
void scanForRightHand(){
  Serial.println("Scanning for RightGlove...");
  while(true){
    peripheral = BLE.available();
    if(peripheral) {
      if (peripheral.localName() == "RightGlove") {
        Serial.println("RightGlove found!");
        BLE.stopScan();
        break;
```

```
      }

    }
    delay(10);
  }
}
```

The own sensor data can easily be collected just by reading it, and the data from the right glove can simply be read from the characteristic. To safe the data into tables, there is also a function to send the whole data to the Serial Monitor:

```
void sendToPC(float* leftValues, float* rightValues) {
  if (!leftValues || !rightValues) return;
  for (int i = 0; i < 6; i++) {
    Serial.print(leftValues[i], 2);
    Serial.print(",");
  }
  for (int i = 0; i < 5; i++) {
    Serial.print(rightValues[i], 2);
    Serial.print(",");
  }
  Serial.println(rightValues[5], 2);
}
```

4.1.2 App Test

4.2 3D Print

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4.3 AI Model Training

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4.4 App Development

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5 Content C

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6 Case Study / Evaluation

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7 Project Management

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7.1 Planning

7.2 Evaluation

7.3 Timesheet

7.3.1 Student A

Datum	Uhrzeit von/bis	Arbeits-Stunden	Arbeitsprozess Thema, Tätigkeit, Arbeitsschritte, Arbeitsablauf, evtl. Koordination externer Partner	Anmerkungen, Hilfsmittel und Hilfestellungen durch Betreuungslehrer/in bzw. externe Berater

…….……..... …….……

Ort und Datum Vorname Nachname

7.3.2 Student B

Datum	Uhrzeit von/bis	Arbeits-Stunden	Arbeitsprozess Thema, Tätigkeit, Arbeitsschritte, Arbeitsablauf, evtl. Koordination externer Partner	Anmerkungen, Hilfsmittel und Hilfestellungen durch Betreuungslehrer/in bzw. externe Berater

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Ort und Datum Vorname Nachname

8 Future Work

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9 Related Work

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10 Conclusion

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List of Figures

Figure 1: Some Description..... **Fehler! Textmarke nicht definiert.**

Bibliography

[1] D. Knuth, The Art of Computer Programming, 1968.

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Kapitel	Name	Bearbeiter/in
	Abstract	Zauner / Stepany
	Zusammenfassung	Zauner / Stepany
1.1	Problem Setting	Zauner